Lagrangian and Hamiltonian Mechanics

Analytical Mechanics

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Definition

When studying the motion of a system it is not enough to solve the differential equations arising from Newton's second law, in general, it is necessary to take in account the **constraints** *that limit the motion of the system. If the conditions of constraint can be expressed as equations connecting the coordinates of the particles (and possibly the time) having the form*

$$f(\boldsymbol{r}_{1/O}, \boldsymbol{r}_{2/O}, \ldots, t) = 0,$$

then the constraints are said to be **holonomic** *and a system whose constraint equations, if any, are all holonomic is called a* **holonomic system**. *Constraints not expressible in this fashion are called* **nonholonomic** *and a system containing at least one such constraint is called a* **nonholonomic system**. *Constraints are further classified according to whether the equations of constraint contain the time as an explicit variable (***rheonomous***) or are not explicitly dependent on time (***scleronomous***). Similarly, the coordinates* $\boldsymbol{r}_{\alpha/O}$ *are classified according to whether they are not explicitly dependent on time* $\partial \boldsymbol{r}_{\alpha/O} \partial t = 0$ *(***scleronomous coordinates***) or contain the time as an explicit variable (***rheonomous coordinates***).*

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Remark

Constraints introduce two types of difficulties in the solution of mechanical problems:

I) *First, the coordinates r_i are no longer independent, since they are connected by the equations of constraint; hence the equations of motion are now coupled differential equations instead of independent ones.*

II) *Second, the forces of constraint are not furnished a priori, and they are among the unknowns of the problem and must be obtained from the solution we seek.*

Indeed, imposing constraints on a system is simply another method of stating that there are forces present in the problem that cannot be specified directly but are known rather in terms of their effect on the motion of the system.

Definition

For holonomic systems, the first difficulty is solved by introducing new coordinates as follows. A system of N particles, free from constraints, has $3N$ independent coordinates or **degrees of freedom**. *Consider a holonomic system with K equations of constraint*

$$f_k(\boldsymbol{r}_{1/O}, \boldsymbol{r}_{2/O}, \ldots, \boldsymbol{r}_{N/O}; t), \quad k = 1, \ldots, K.$$

In this case we can use the equations of constraint to eliminate K of the $3N$ coordinates to get $M = 3N - K$ degrees of freedom. With such a reduction we can always (theoretically) introduce a new set of M independent variables $q_1, \ldots, q_M$ —sometimes called a **minimal** *set— in terms of which the old coordinates $\boldsymbol{r}_{1/O}, \ldots, \boldsymbol{r}_{N/O}$ are expressed by equations of the form*

$$\begin{aligned} \boldsymbol{r}_{1/O} &= \boldsymbol{r}_{1/O}(q_1, \ldots, q_M; t) \\ &\vdots \\ \boldsymbol{r}_{N/O} &= \boldsymbol{r}_{N/O}(q_1, \ldots, q_M; t) \end{aligned}$$

containing the constraints in them implicitly. These are **transformation** *equations from the set $(\boldsymbol{r}_{j/O})$ to the set (q_j) of* **generalised coordinates**. *It is always assumed that we can also transform back from the (q_j) to the $\boldsymbol{r}_{j/O}$.*

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Example 1.1

Consider a particle constrained to move in a curve $y = h(x)$ in the xy-plane. Find the degrees of freedom, constraints and possible choices of generalised coordinates.

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Example 1.2

Discuss the degrees of freedom, constraints and admissible choices of generalised coordinates for each of the following systems.

i) $N = 2$ *particle system in space with no constraints.*

ii) $N = 2$ *particles in a plane with no constraints.*

iii) $N = 2$ *particles in a plane connected by a rigid rod.*

iv) *Rigid body in a plane.*

v) *Two rigid bodies in a plane.*

vi) *Two rigid bodies in a plane attached by a pin joint.*

vii) *Rigid body in space.*

viii) *Two rigid bodies connected by a ball of socket joint.*

Remark

Suppose that the position of a particle P is represented by an inertial frame $\mathcal{I}$ so that

$$\boldsymbol{r} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}_{\mathcal{I}},$$

when we write $\boldsymbol{r}_{P/O}$ in terms of another set of generalised coordinates $r_i = r_i(\mathbf{q}_j)$, in such case we would have

$${}^{\mathcal{I}}\mathbf{v}_{P/O} = \dot{r}_i e_i = \dot{q}_j \frac{\partial r_i}{\partial q_j} e_i,$$

so that now, the velocity ${}^{\mathcal{I}}\mathbf{v}_{P/O}$ is a function of the **generalised velocities** *$(\dot{q}_j)$ and the generalised coordinates (q_j) and similarly, ${}^{\mathcal{I}}\mathbf{a}_{P/O}$ will be a function of $(q_j, \dot{q}_j, \ddot{q}_j)$. This idea is readily generalised to a system of N-particles and the new set of n generalised coordinates used to describe the such system form what is known as the* **configuration space** *$(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n)$*

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Remark

A common type nonholonomic constraints are those on the rate of change of the generalised coordinates $\dot{q}_i$, i.e., the generalised velocities. Such constraints have the general form

$$\sum_{i=1}^{M} \xi_{\ell,i} \dot{q}_i + \xi_{l,t} = 0, \quad l = 1, \ldots, S$$

where $\xi_{l,i}(q_1, \ldots, q_M; t)$ and $\xi_{l,t}(q_1, \ldots, q_M; t)$ are functions of the generalised coordinates and S is the number of nonholonomic constraints.

Example 1.3

Consider a disc rolling on the horizontal xy-plane constrained to move so that the plane of the disc is always vertical. The coordinates used to describe the motion might be the usual Cartesian coordinates of the centre of the disc, an angle of rotation ϕ about the axis of the disc, and an angle θ between the axis of the disc and say, the x-axis (cf. Fig 1). Find and discuss the equations of constraint.

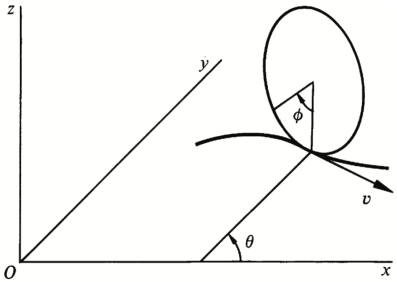


Figure 1: Vertical disc rolling on a horizontal plane

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Example 1.4

Figure 2 shows a simple nonholonomic system. It consists of two particles, P and Q, connected by a massless rod of length l, lying flat on an icy surface. Each particle rests on a massless blade such that it can only move along the direction of the blade. Discuss the equations of constraint.

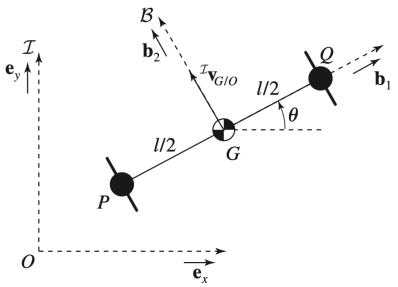


Figure 2: Simple nonholonomic system. The particles P and Q are constrained to move along the directions of the blades

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Definition

Suppose the configuration of a system is specified by the n generalised co-ordinates q_i. For holonomic systems we can always write the equations of constraint in the form of K independent algebraic equations of the form

$$\phi_k(q_1, \ldots, q_n; t) = 0 \quad k = 1, \ldots, K$$

where K is the total number of constraints.

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Remark

To surmount the second difficulty, we will formulate mechanics in such a way the forces of constraint disappear and we are only left with applied forces. The next section develops such a formulation.

Historical Notes



Jean Le Rond d'Alembert
1717-1783

Jean Le Rond d'Alembert (1717-1783) was a French mathematician, physicist and philosopher. Abandoned by his mother, d'Alembert was raised by a foster family and later educated by the arrangement of his father at a nearby church-sponsored school, in which he received instruction in the classics and above-average instruction in mathematics. He travelled little and worked at the Paris Academy of Science and the French Academy all his life. On another level his life was one of great drama as he argued with almost everyone around him. Euler had learnt of d'Alembert's work in around 1743 through letters from Daniel Bernoulli. However relations between Euler and d'Alembert soon took a turn for the worse. A reason for d'Alembert to feel angry with Euler was that he felt that Euler was stealing his ideas and not giving him due credit. Perhaps his most famous scientific work, entitled *Traité de dynamique*, shows his appreciation that a revolution was taking place in the science of mechanics—the formalisation of the principles stated by Newton into a rigorous mathematical framework. The philosophy to which d'Alembert subscribed, however, refused to acknowledge the primacy of a concept as unclear and arbitrary as "force," introducing a certain awkwardness to his treatment and perhaps causing him to overlook the important principle of conservation of energy. Later, d'Alembert produced a treatise on fluid mechanics, a paper dealing with vibrating strings (in which the wave equation makes its first appearance in physics), and a skillful treatment of celestial mechanics. His ideas on limits led him to the test for convergence, known today as d'Alembert's ratio test.

Theorem 1.5

Consider a system of N particles for which **D'Alembert's principle**

$$\sum_{j=1}^{N} \mathbf{F}_j^{(c)} \cdot \frac{\partial {}^{\mathcal{I}}\mathbf{r}_j}{\partial q_i} = 0$$

holds. Then the equation

$$\sum_{j=1}^{N} \left(\mathbf{F}_j^{(a)} - m_j \frac{\mathrm{d}}{\mathrm{d}t} \left({}^{\mathcal{I}}\mathbf{v}_j \right) \right) \cdot \frac{\partial {}^{\mathcal{I}}\mathbf{r}_j}{\partial q_i} = 0$$

yields the equation of motion for the system.

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Example 1.6

Here are some examples of systems for which D'Alembert's principle holds:

a) *Mass constrained to slide through a surface.*

b) *Pendulums.*

c) *Rigid interconnections.*

d) *Rope pulley-system (Atwood's machine).*

e) *Rolling without slipping.*

f) *Mass moving on a wire.*

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Example 1.7

Show that d'Alembert's principle holds for a simple pendulum and use this fact to derive the equations of motion.

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Theorem 1.8

For a multi-particle system, consider the transformation equations

$$\boldsymbol{r}_j(q_1, \ldots, q_n), \quad j = 1, \ldots, N$$

and define the so-called **projection vectors** $\boldsymbol{\gamma}_{ij} \triangleq \partial \boldsymbol{r}_i / \partial q_j$. *Then*

$$\boldsymbol{\gamma}_{ij} = \frac{\partial \boldsymbol{r}_i}{\partial q_j} = \frac{\partial \mathbf{v}_i}{\partial \dot{q}_j}.$$

The elements $\partial \mathbf{v}_i / \partial \dot{q}_j$ *are called* **velocity coefficients**. *With these definitions one can write*

$$\delta \boldsymbol{r}_i = \boldsymbol{\gamma}_{ij} \delta q_j.$$

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Definition

We define the **generalised forces** Q_i *to be the numbers*

$$Q_i = \mathbf{F}_j^{(a)} \cdot \boldsymbol{\gamma}_{ji}.$$

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Theorem 1.9 (Lagrange's Equations)

If $(q_1, \ldots, q_n)$ is a minimal set of generalised coordinates, then d'Alembert's principle of virtual work can be rewritten as

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i} = Q_i$$

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Example 1.10

For a spring mass system, find the generalised forces.

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Definition

Let us now define the **natural** *(or* **standard***)* **Lagrangian** *of a system to be the quantity*

$$L = T - V$$

where T *and* V *are the total kinetic and potential energy of the system, respectively.*

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Notation

We will write $(q_1, \ldots, q_n) = \mathbf{q}$ *so that* $(\dot{q}_1, \ldots, \dot{q}_n) = \dot{\mathbf{q}}$ *and* $(\ddot{q}_1, \ldots, \ddot{q}_n) = \ddot{\mathbf{q}}$. *In this notation*

$$\frac{\partial L}{\partial \mathbf{q}} \quad \textit{means} \quad \frac{\partial L}{\partial q_i}\hat{\imath} \equiv \left(\frac{\partial L}{\partial q_1}, \ldots, \frac{\partial L}{\partial q_n}\right),$$

with similar meanings for $\partial L/\partial \dot{\mathbf{q}}$ *and* $\partial L/\partial \ddot{\mathbf{q}}$.

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Theorem 1.11 (Lagrange's Equations in Standard Form)

Lagrange's equations can be rewritten as

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\mathbf{q}}} - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q}^{(nc)}$$

where for nonconservative forces $\mathbf{F}_j^{(nc)}$ *we define* $Q_i^{(nc)} = \mathbf{F}_j^{(nc)} \cdot \boldsymbol{\gamma}_{ji}$.

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Definition

A **monogenic** *system is a holonomic system in which all the generalised applied forces arise from a velocity-dependent potential* $V(\mathbf{q}, \dot{\mathbf{q}})$ *via*

$$\mathbf{Q} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial V}{\partial \dot{\mathbf{q}}} - \frac{\partial V}{\partial \mathbf{q}}.$$

In this case V *is called a* **generalised potential**.

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Theorem 1.12 (Euler-Lagrange Equations)

For a monogenic system, Lagrange's equations can be written as

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\mathbf{q}}} - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{0},$$

where $L = T - V$, with V a generalised potential.

Example 1.13

Show that the **electromagnetic Lagrangian**

$$L = \frac{1}{2}m\mathbf{v}^2 + q(\mathbf{A}\cdot\mathbf{v} - \varphi)$$

arising from the generalised potential

$$V = q(\varphi - \mathbf{A}\cdot\mathbf{v})$$

gives rise to the **Lorentz equation**

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B})$$

by means of the Euler-Lagrange equations, where the potentials $\mathbf{A}, \varphi$ *are defined by*

$$\mathbf{E} = -\nabla\varphi - \frac{\partial\mathbf{A}}{\partial t} \quad \textit{and} \quad \mathbf{B} = \nabla\times\mathbf{A}.$$

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Definition

Quantities that remain constant during the motion of a system are called **integrals of the motion**.

Definition

A one-parameter family of transformations

$$q_i^{(s)} = q_i^{(s)}(q_j(t), t, s), \quad t^{(s)} = t^{(s)}(q_j(t), t, s)$$

such that $q_i^{(0)} = q_i, t^{(0)} = t$, *is said to be a family* **strict symmetries of the Lagrangian** *if*

$$\int_{t_0}^{t_1} \mathrm{d}t L(q_i, \mathrm{d}q_i/\mathrm{d}t, t) - \int_{t_0}^{t_1} \mathrm{d}t \frac{\mathrm{d}t^{(s)}}{\mathrm{d}t} L\left(q_i^{(s)}(t), \frac{\mathrm{d}q_i/\mathrm{d}t}{\mathrm{d}t^{(s)}/\mathrm{d}t}, t^{(s)}(t)\right) = 0,$$

for all $t_0, t_1, s \in \mathbb{R}$, *and a family of* **Noether-Bessel-Hagen** *(or* **variational***)* **symmetries** *if*

$$\int_{t_0}^{t_1} \mathrm{d}t L(q_i, \mathrm{d}q_i/\mathrm{d}t, t) - \int_{t_0}^{t_1} \mathrm{d}t \frac{\mathrm{d}t^{(s)}}{\mathrm{d}t} L\left(q_i^{(s)}(t), \frac{\mathrm{d}q_i/\mathrm{d}t}{\mathrm{d}t^{(s)}/\mathrm{d}t}, t^{(s)}(t)\right) = F(q_i(t_1), t_1, s) - F(q_i(t_2), t_2, s)$$

for some function $F = F(q_i, t, s)$.

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Definition

The quantity J defined as

$$J = \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i - L$$

is called **Jacobi's integral**.

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Lemma

For a monogenic system with a velocity independent potential V and a kinetic energy that is a homogeneous function of order two in the generalised velocities $\dot{q}_1, \ldots, \dot{q}_n$, Jacobi's integral becomes $J = T + V$.

Theorem 1.14 (Noether's First Theorem for Point Transformations)

If a one-parameter family of transformations

$$q_i^{(s)} = q_i^{(s)}(q_j(t), t, s), \quad t^{(s)} = t^{(s)}(q_j(t), t, s)$$

define a family of Noether-Bessel-Hagen symmetries of a Lagrangian L and satisfy the Euler-Lagrange equations, then for

$$\xi = \left.\frac{\partial t^{(s)}}{\partial s}\right|_{s=0}, \quad \eta_i = \left.\frac{\partial q_i^{(s)}}{\partial s}\right|_{s=0}, \quad G = \left.\frac{\partial F}{\partial s}\right|_{s=0}, \quad J = \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i - L$$

the quantity

$$N = \frac{\partial L}{\partial \dot{q}_i}\eta_i - J\xi - G,$$

called the **Noether charge**, *is an integral of the motion. Conversely, if the family of transformations is such that*

$$\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}\eta_i}{\mathrm{d}t} + \frac{\partial L}{\partial t}\xi + \frac{\mathrm{d}\xi}{\mathrm{d}t}\left(L - \frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\mathrm{d}G}{\mathrm{d}t},$$

then the family define a family of Noether-Bessel-Hagen symmetries of L.

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Definition

Given a set of generalised coordinates $(q_1, \ldots, q_n)$ *and a Lagrangian* $L = L(q_i, \dot{q}_i, t)$, q_k *is said to be an* **ignorable coordinate** *if* $\partial L / \partial q_k = 0$.

Example 1.15

Let $q_1, \ldots, q_n$ be a set of generalised coordinates and let q_k be an ignorable coordinate of a Lagrangian $L(q_i, \dot{q}_i, t)$, then

$$\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}\eta_i}{\mathrm{d}t} + \frac{\partial L}{\partial t}\xi + \frac{\mathrm{d}\xi}{\mathrm{d}t}\left(L - \frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\mathrm{d}G}{\mathrm{d}t}$$

for the selection $G = 0, \xi = 0, \eta_i = \delta_{ik}$, so

$$N = \frac{\partial L}{\partial \dot{q}_i}\delta_{ik} = \frac{\partial L}{\partial \dot{q}_k}$$

is conserved.

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Definition

The **generalised** *(or* **canonical***)* **momentum** $\mathbf{p}$ *is the quantity defined as*

$$\mathbf{p} = \frac{\partial L}{\partial \dot{\mathbf{q}}}.$$

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Example 1.16

Let $q_1, \ldots, q_n$ be a set of generalised coordinates and let $\partial L/\partial t = 0$, then

$$\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}\eta_i}{\mathrm{d}t} + \frac{\partial L}{\partial t}\xi + \frac{\mathrm{d}\xi}{\mathrm{d}t}\left(L - \frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\mathrm{d}G}{\mathrm{d}t}$$

for the selection $G = 0, \xi = 1, \eta_i = 0$, so

$$N = J\xi = J$$

is conserved.

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Definition

The Lagrangian is called **regular** *if the* **Hessian matrix**

$$(H)_{ij} = \frac{\partial^2 L}{\partial \dot{q}_i \dot{q}_j}$$

has nonzero determinant. A Lagrangian failing such condition is said to be **singular**.

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Definition

Let us define the **Hamiltonian** *of a system as*

$$H(\mathbf{p}, \mathbf{q}, t) = \mathbf{p} \cdot \dot{\mathbf{q}} - L$$

where $\mathbf{p} = \partial L / \partial \dot{\mathbf{q}}$ *is the canonical momentum.*

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Theorem 1.17

For a Lagrangian of the form

$$L = \frac{1}{2} a_{ij} \dot{q}_i \dot{q}_j - V(\mathbf{q}, t),$$

where the a_{ij} *are functions of* $(\mathbf{q}, t)$ *only, with* $A = (a_{ij})$ *symmetric and* $\det A \neq 0$*, the Hamiltonian is given by*

$$H = \frac{1}{2} b_{ij} p_i p_j + V(\mathbf{q}, t),$$

where $A^{-1} = (b_{ij})$ *is also symmetric.*

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Theorem 1.18 (Hamilton's Canonical Equations)

Let L be a regular Lagrangian. Then the **Hamilton's canonical equations**

$$\frac{\partial H}{\partial \mathbf{q}} = -\dot{\mathbf{p}}, \quad \frac{\partial H}{\partial \mathbf{p}} = \dot{\mathbf{q}}, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}$$

hold and are equivalent to the Euler-Lagrange equations. The variables $(\mathbf{p}, \mathbf{q})$ are said to be **canonical coordinates** *and due to the first two canonical equations, $\mathbf{q}$ and $\mathbf{p}$ are said to be* **canonically conjugate** *to one another.*

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Example 1.19

Find the Hamiltonian for a simple harmonic oscillator and use Hamilton's canonical equations to deduce the equation of motion of the system.

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Theorem 1.20

If $\partial H/\partial t = 0$, then H is an integral of the motion.

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Theorem 1.21

If the Hamiltonian is a sum of the form

$$H(q_i, p_i, t) = h_1(q_1, \ldots, q_k, p_1, \ldots, p_k) + h_2(q_{k+1}, \ldots, q_n, p_{k+1}, p_n, t),$$

for some $1 \leq k \leq n$, then h_1 is a constant of the motion.

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Theorem 1.22 (Poisson's Theorem)

If f and g are differentiable functions defined on the extended phase space, then

$$\frac{\mathrm{d}\{f,g\}}{\mathrm{d}t} = \{f, \mathrm{d}g/\mathrm{d}t\} + \{\mathrm{d}f/\mathrm{d}t, g\}.$$

Hence, if f and g are constants of the motion (that may depend explicitly on the time), then $\{f,g\}$ is also a constant of the motion.